once the gravitational field has completely spread throughout the extra curled-up dimension, it will not diminish with further separation.

From your data you would deduce two things. First, from the fact that the gravitational force diminished in proportion to distance when objects are very close, you'd realize that the universe has *two* space dimensions, not one. Second, from the crossover to a gravitational force that is constant—the result known from hundreds of years of previous experiments—you'd conclude that one of these dimensions is curled up, with a size about equal to the distance at which the crossover takes place. And with this result, you'd overturn centuries, if not millennia, of belief regarding something so basic, the number of space dimensions, that it seemed almost beyond questioning.

Although I set this story in a lower-dimensional universe, for visual ease, our situation could be much the same. Hundreds of years of experiments have confirmed that gravity varies inversely with the square of distance, giving strong evidence that there are three space dimensions. But as of 1998, no experiment had ever probed gravity's strength on separations smaller than a millimeter (today, as mentioned, this has been pushed to a tenth of a millimeter). This led Savas Dimopoulos, of Stanford, Nima Arkani-Hamed, now of Harvard, and Gia Dvali, of New York University, to propose that *in the braneworld scenario extra dimensions could be as large as a millimeter and would still have been undetected.* This radical suggestion inspired a number of experimental groups to initiate a study of gravity at submillimeter distances in hopes of finding violations of the inverse square law; so far, none have been found, down to a tenth of a millimeter. Thus, even with today's state-of-the-art gravity experiments, *if we are living within a three-brane, the extra dimensions could be as large as a tenth of a millimeter, and yet we wouldn't know it.*

This is one of the most striking realizations of the last decade. Using the three nongravitational forces, we can probe down to about a billionth of a billionth (10^{-18}) of a meter, and no one has found any evidence of extra dimensions. But in the braneworld scenario, the nongravitational forces are impotent in searching for extra dimensions since they are trapped on the brane itself. Only gravity can give insight into the nature of the extra dimensions, and, as of today, the extra dimensions could be